Private Higher School of Engineering and Technologies (ESPRIT)

Integration Project Report

Project Title:

Intelligent platform for managing training programs and professional certifications

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General Introduction

IN the digital era, continuous learning and skill development have become essential to meet the demands of a constantly evolving labor market. Both organizations and individuals face the need to acquire new skills to adapt to technological, economic, and social transformations.

In this context, training platforms play a central role in providing access to knowledge and enhancing professional competencies. However, traditional training management systems often have limitations in terms of personalization, efficiency, and intelligent data utilization. Learners sometimes struggle to identify the most suitable courses for their needs, while organizations face challenges in managing and monitoring training programs.

To address these issues, this integration project aims to design and develop an intelligent platform for managing professional training and certifications. This platform will centralize the management of courses, users, enrollments, and assessments while integrating an intelligent recommendation module based on Machine Learning techniques. The goal is to provide a more efficient, personalized, and scalable digital environment that meets the expectations of both learners and organizations.

This report presents the various stages of project design and planning, focusing on needs analysis, system modeling, adopted architecture, and development planning. It this provides a solid foundation for the technical implementation of the platform.

Chapter 1

Project Framework Presentation

Introduction

In the digital era, continuous learning and skill development are essential to meet the demands of a constantly evolving labor market. Learners struggle to find suitable courses, and organizations face challenges in monitoring training and certifications.

This project proposes an intelligent platform for managing professional training programs and certifications, centralizing management of users, courses, exams, skills, sessions and equipment, and partners, and providing personalized recommendations using Machine Learning.

1.1 General framework of the work

The platform optimizes training pathways, provides tailored course recommendations, and offers a scalable, efficient solution for learners and organizations.

1.1.1 Problem Statement

Learners often struggle to find training programs that match their skills, while organizations face challenges in effectively monitoring training pathways and certifications. Therefore, it is necessary to provide a solution capable of managing training programs while automatically recommending the most relevant courses. This recommendation will be personalized based on each learner's profile and skills.

1.2 Study of Existing Solutions

1.2.1 Description of the Existing Situation

Currently, several platforms are used for managing training programs and professional certifications. The most well-known are Moodle, Coursera, and Udemy, each with its own characteristics.

Moodle: a widely used open-source platform in educational institutions and companies. It allows the creation of online courses, user management, assessments, and discussion forums.

Coursera: a commercial platform offering online courses and certificates issued by partner universities or institutions. It provides an intuitive interface and a large course catalog.

Cisco: a commercial platform for technical training and certifications in networking and IT infrastructure

1.2.2 Critique of the Existing Situation

Although existing platforms such as Moodle, Coursera and Cisco provide useful features for managing training programs and certifications, they present several limitations that reduce their effectiveness and adaptability to the current needs of learners and organizations.

Features / Platforms	Cisco	Moodle	Coursera	Skill2Job
Online Courses	✓	✓	✓	✓
Presential Courses	✗	✗	✗	✓
Professional Certifications	✓	✗	✓	✓
Industry Recognition	✓	✗	✓	✓
Custom Course Creation	✗	✓	✗	✓
Exams / Quizzes	✓	✓	✓	✓
Automatic Certificate Generation (PDF)	✓	□	✓	✓
Progress Tracking	✓	✓	✓	✓
Intelligent Recommendations	✗	✗	✓	✓

Table 1 : comparative analysis of existing E-Learning platforms

In summary, the proposed solution offers an intelligent, personalized and user-centered approach that addresses the shortcomings of existing platforms and brings significant added value to professional training.

1.3 Methodological Choices and Adopted Formalism

1.3.1 Comparative Study between Classical and Agile Methods

Classical methods, such as the Waterfall model, follow a linear and sequential process where each phase must be completed before moving to the next one. This lack of flexibility makes them less suitable for projects in which requirements may evolve over time

In contrast, Agile approaches rely on iterative and incremental cycles that promote continuous feedback, close collaboration with stakeholders, and frequent delivery of functional product increments.

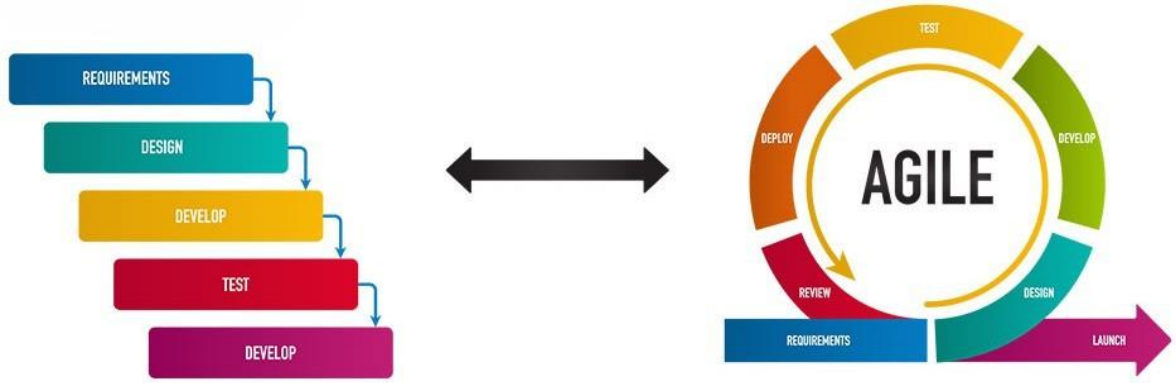


Figure 1: Comparison between classical and agile methods

1.3.1.1 Why Scrum?

Scrum was chosen for its ability to manage complex and evolving projects through short development cycles (sprints) and regular reviews with stakeholders. This approach improves alignment with expectations, facilitates early feedback, and reduces the risk associated with changing requirements.

1.3.1.2 Scrum Framework

The Scrum framework organizes work into fixed-length iterations called *sprints* (we propose 2-week sprints). Each sprint delivers a potentially shippable increment. Key Scrum artifacts include the Product Backlog, the Sprint Backlog and the Definition of Done. Ceremonies comprise Sprint Planning, Daily Scrum, Sprint Review and Sprint Retrospective. These practices promote transparency, inspection and adaptation throughout the development process.

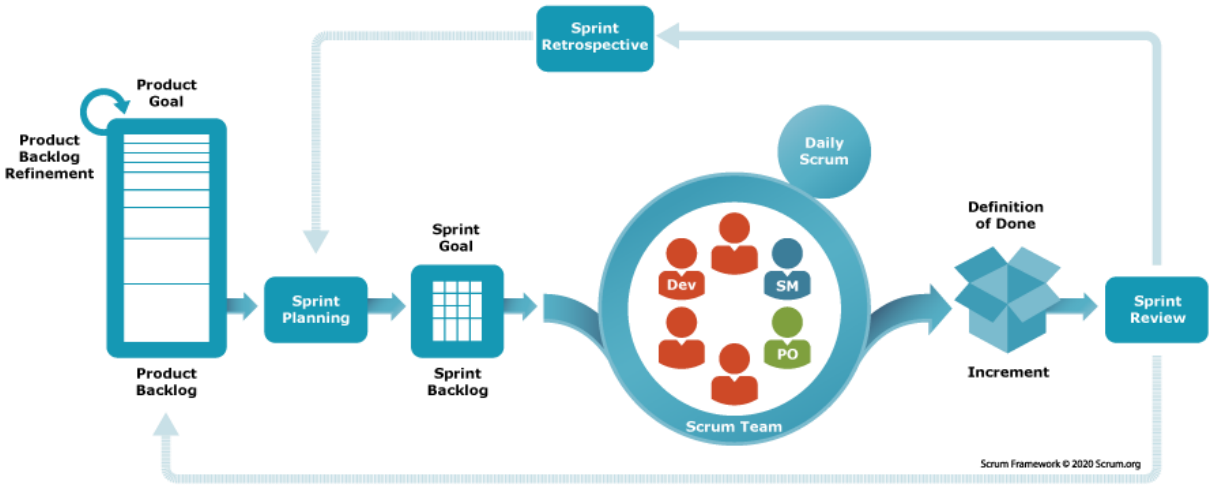


Figure 2 : Scrum Framework: roles, artifacts and ceremonies

1.3.1.3 The Scrum Team

- **Product Owner**: responsible for defining priorities and maintaining the product backlog.
- **Development Team**: implements features and ensures the quality of deliverables.
- **Scrum Master**: facilitates the Scrum process, removes impediments and protects the team from external disruptions.

Conclusion

In summary, Scrum and UML have been selected to support iterative delivery and clear system modeling. Chapter 2 will present detailed architecture, microservices design and the project plan, including sprint schedules and the CI/CD pipeline.

Chapter 2

Planning and System Architecture

Introduction

This chapter details the planning and the system architecture of the project. It describes the stakeholders and requirements, presents UML models used for specification, and defines the technical architecture at both high and component level. We also present the design of the Machine Learning recommendation module and the DevOps strategy (CI/CD, containerization, orchestration). Finally, a sprint plan and the development environment are proposed to support implementation.

2.1 Stakeholders and Actors Identification

The main stakeholders and actors involved in the platform are:

- **Visitor**: a non-authenticated user who can consult the course catalog and register on the platform.
- **Learner**: an authenticated user who can enroll in courses, manage their profile, follow learning progress, take assessments, download certificates, receive notifications, and obtain personalized course recommendations.
- **Instructor:** responsible for creating and managing course content“, managing assessments, and monitoring learners’ progress through analytics dashboards.”
- **Administrator**: responsible for managing users (learners and instructors), roles and permissions, and accessing global reports for platform supervision.
- **Partner**: a company representative who manages the organization’s profile, publishes internship or job offers, reviews student applications, and selects candidates for recruitment opportunities.
- **Recommendation Engine**: a system component that generates personalized course recommendations based on learners’ profiles, skills, and learning history.

2.2 Requirements Analysis

2.2.1 Functional Requirements

Key functional requirements include:

- **User Management:** registration, authentication, profiles, roles
- **Training Courses Management:** CRUD operations for courses and modules
- **Exams Management:** create quizzes/exams, grading, certificate issuance
- **Trainer Recruitment Management**: submission, review, AI-based evaluation, and approval or rejection of trainer applications.
- **Partners Management:** manage companies, internships, and collaborations
- **Sessions and Rooms Management**: manage equipment based on sessions schedule

2.2.2 Non-Functional Requirements

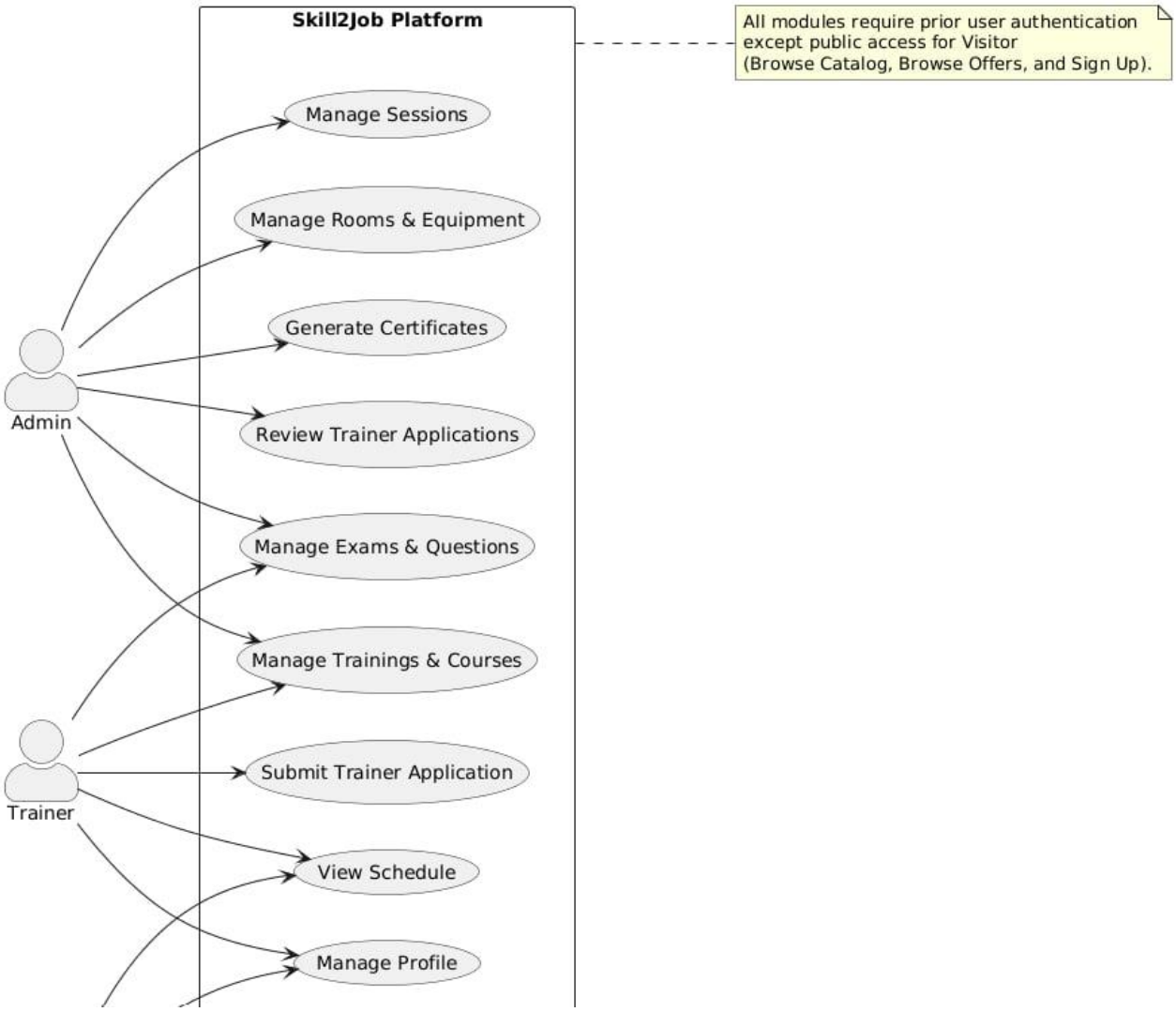
Non-functional constraints and expectations:

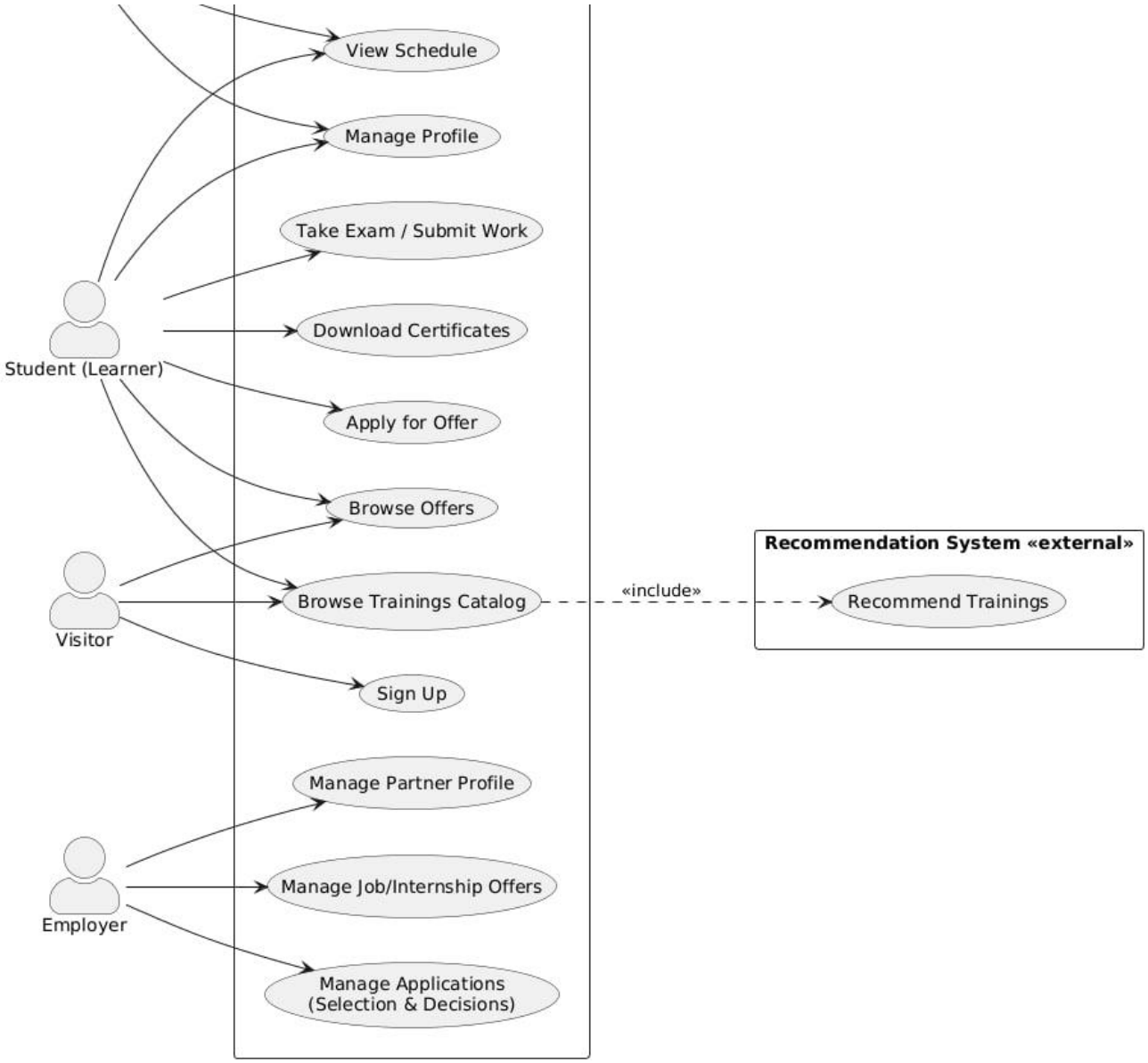
- **Scalability**: Stateless backend (JWT) , Multiple API instances + Load Balancer ,Database indexing & pagination .
- **Availability**: Redundant backend instances ,Database backups ,Health monitoring
- **Security & Privacy**: HTTPS communication , JWT authentication + Role-based access ,Password hashing .
- **Performance:** Optimized SQL queries (indexes) ,Pagination , Production build (Angular).

2.3 System Modeling (UML)

2.3.1 Use Case Diagram

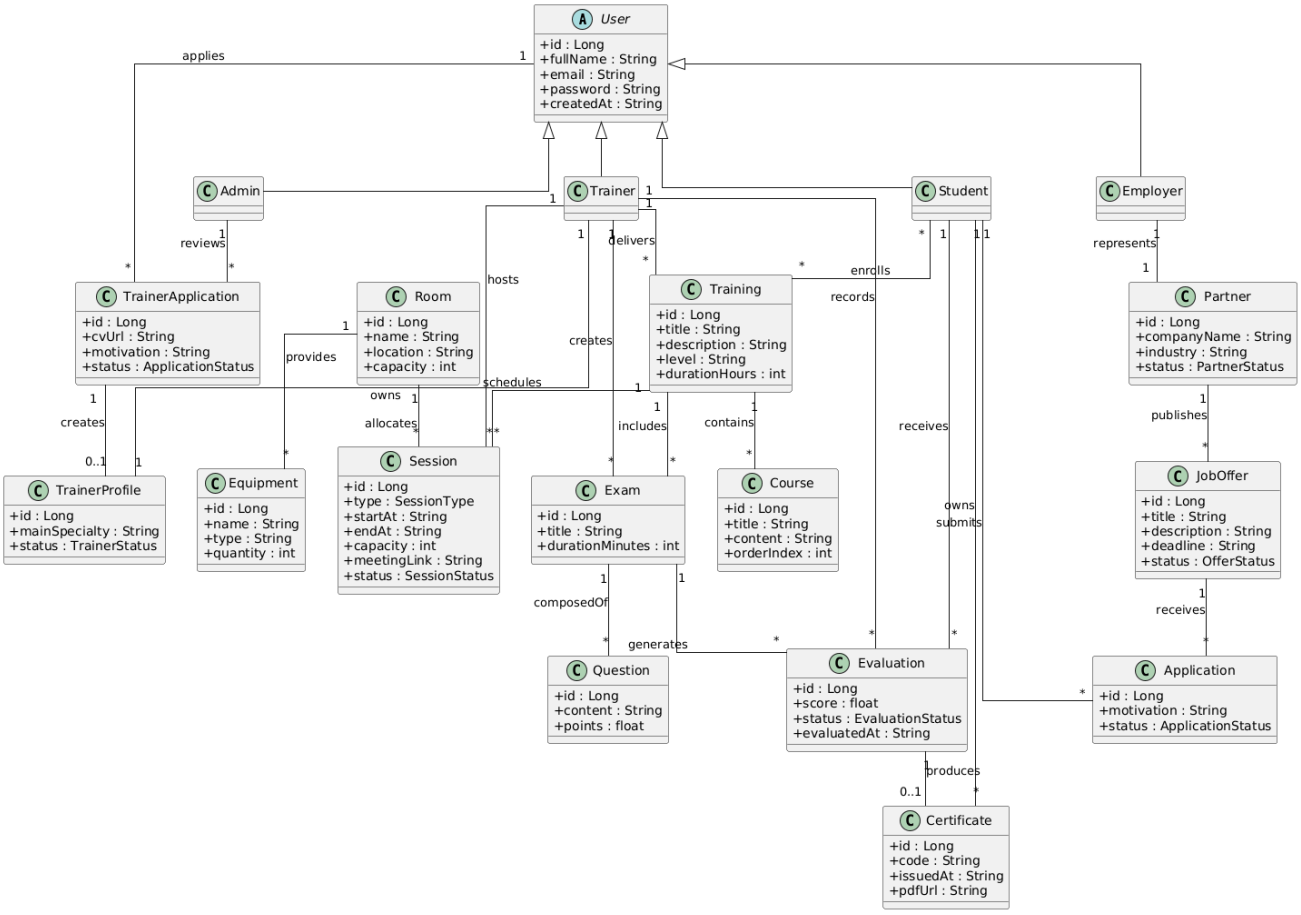
A high-level use case diagram will show interactions between actors and main use cases




View Schedule
Manage Profile
Take Exam / Submit Work
Download Certificates
Student (Learner)
Apply for Offer
Browse Offers
Recommendation System «external»
Visitor
Browse Trainings Catalog
«include»
Recommend Trainings
Sign Up
Manage Partner Profile
Employer
Manage Job/Internship Offers
Manage Applications
(Selection & Decisions)

2.1.1 Class Diagram

Define domain classes:



2.1 System Architecture Design

2.1.1 Global Architecture Overview

The system adopts microservice architecture. Services communicate via REST for synchronous operations and via a message broker (RabbitMQ or Kafka) for asynchronous events. Cross-cutting layers include an API Gateway, service discovery, data storage per service, and centralized observability.

2.1.2 Microservices Architecture

Microservices Identification

Suggested services:

- Users Service
- Training Courses Service
- Sessions and Equipment Service
- Partners Service
- Exams Service
- Trainer Recruitment Management

2.2 Project Planning

2.2.1 Product Backlog

Maintain a prioritized backlog with epics and user stories

Module	User Stories	Estimation (weeks)
User Management	Registration, authentification, rôles, profile management	2
Training Courses	CRUD courses/modules, materials upload	2
Exams Management	Create quizzes/exams, grading, certificates	2
Skills Management	Map skills to learners and courses, progress tracking	2
Partner Management	Company profiles, internships, collaborations	2
Sessions and Equipment Management	CRUD Equipment, schedule sessions, track availability, allocate classrooms	2
Recommandation Engine	Personalized course suggestions	2

Table 2: system modules, user stories and time estimation

2.1 Development Environment

2.1.1 Hardware Environment

Developers will use modern developer machines; CI uses cloud runners; staging/prod uses cloud-managed infrastructure.

2.1.2 Software Environment

- Backend: Spring Boot.
- Frontend: Angular.
- Database : MySQL.

Conclusion

This chapter defined a modular microservices architecture, a practical ML integration design and a DevOps strategy to enable continuous delivery. The proposed planning and technical choices balance short-term delivery (MVP) with long-term maintainability and scalability.

The next chapter will detail the API contracts, data models and implementation specifics for microservices.